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Research Article

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Determining the minimum number of steps required for reliable kinetic, kinematic, gait variability, and $\dot{V}iO_2$ parameters in outdoor running

Johnny Padulo[1,*], Raffaele Scurati[1], Simone Baldo[1], Fabio Esposito[1], Leonardo Alexandre Peyré-Tartaruga[2,3]

[1]Department of Biomedical Sciences for Health, Università degli Studi di Milano, 20133 Milan, Italy
[2]Unidad Académica de Biofísica, Facultad de Medicina, Universidad de la República, 11800 Montevideo, Uruguay
[3]Human Locomotion Laboratory (LocoLab), Department of Public Health, Experimental and Forensic Medicine, University of Pavia, 27100 Pavia, Italy

*Corresponding author. Department of Biomedical Sciences for Health, Università degli Studi di Milano, 20133 Milan, Italy.
E-mail: johnny.padulo@unimi.it

Abstract

Introduction: Outdoor running gait analysis using wearable sensors requires establishing minimum step requirements for reliable parameter measurement.

Sources of data: Fourteen trained runners completed 10-minute trials at critical speed. RunScribe™ captured kinetic/kinematic data at 500 Hz. Convergence analysis determined minimum steps when cumulative means stabilized within ±2% for ≥10 consecutive steps.

Areas of agreement: Most parameters achieved excellent reliability (ICC > 0.900). Metabolic parameters ($\dot{V}iO_2$: 108 ± 32 steps) and step rate (124 ± 36 steps) converged fastest, while kinematic variables required moderate sampling (stride length: 152 ± 48 steps; contact time: 198 ± 52 steps).

Areas of controversy: Complex biomechanical metrics showed substantial variability: leg stiffness (k_{leg}: 342 ± 78 steps; k_{vert}: 298 ± 71 steps) and gait variability (246 ± 12 steps) required substantially more data, challenging brief field assessment feasibility.

Growing points: This study establishes parameter-specific sampling protocols for outdoor gait analysis, demonstrating that different measurement domains require tailored data collection strategies to optimize ecological validity.

Areas timely for developing research: Future research should examine how speed, terrain, and environmental conditions affect sampling requirements. Adaptive protocols and machine learning approaches offer promising opportunities for optimizing data collection efficiency.

Keywords: critical speed; footstep analysis; human locomotion; ecological validity; endurance runners

Introduction

Gait analysis in walking [1], racewalking [2] and running [3–5] has become increasingly important in sports science, clinical biomechanics, and injury prevention research. Accurate quantification of kinetic [6], kinematic [7], gait variability [8–10], and metabolic parameters [11, 12] provides valuable insights into running performance, efficiency, and potential injury risk factors [13]. However, the reliability of these measurements depends critically on obtaining sufficient data samples, particularly when conducting assessments in outdoor [14], real-world environments where conditions are more variable than controlled laboratory settings [15]. Recent advances in wearable sensor technology have enabled sophisticated biomechanical analysis beyond laboratory confines, with devices such as RunScribe™ representing a substantial breakthrough in field-based gait assessment [16]. RunScribe™ is a lightweight, wireless foot pod system that incorporates tri-axial accelerometers, gyroscopes,

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and magnetometers to capture comprehensive kinetic and kinematic data during running [17]. These inertial measurement units (IMUs) can quantify multiple gait parameters, including ground contact time, flight time, cadence, pronation velocity, impact forces, and foot strike patterns with high temporal resolution [18]. The device's ability to continuously monitor biomechanical variables during prolonged outdoor running sessions makes it particularly valuable for ecological validity studies, as it eliminates the spatial constraints (limited capture volume of typically 5–10 meters) and temporal constraints (restricted trial duration due to capture space limitations) inherent in traditional laboratory-based motion capture systems [19]. While natural trail environments present the highest ecological validity, a standardized outdoor track provides a controlled outdoor setting that balances ecological validity with experimental control. This intermediate approach between laboratory treadmills and variable natural terrain allows systematic investigation of sampling requirements while maintaining outdoor running characteristics such as natural environmental conditions, psychological factors, and unrestricted movement patterns [20].

Traditional laboratory-based gait analysis, while offering precise environmental control [14], represents a different context as an ecological approach and does not fully capture the complexity and natural variability of outdoor running [6]. Factors such as terrain variation, wind resistance, temperature fluctuations, and the psychological aspects of running in natural environments can substantially influence biomechanical [21] and physiological responses [22]. Consequently, there is growing interest in conducting gait analysis during outdoor running to obtain more ecologically valid data that better represents actual running conditions.

A fundamental challenge in outdoor gait analysis is determining the minimum number of steps required to achieve reliable and representative measurements [23]. While previous studies have established protocols for laboratory-based assessments [15], the increased variability inherent in outdoor conditions necessitates a re-evaluation of sampling requirements. Insufficient data collection leads to unreliable results and poor repeatability. In contrast, excessive data collection is inefficient and unnecessarily time-consuming for establishing baseline steady-state gait parameters at a given intensity. The parameters of interest in running analysis span multiple domains: kinetic variables reflect the forces generated during ground contact, kinematic parameters describe the movement patterns of body segments, gait variability measures quantify the natural fluctuations in movement patterns, and oxygen consumption ($\dot{V}O_2$) provides insight into metabolic demands. Each parameter type has different reliability characteristics and therefore requires different minimum sampling durations for accurate assessment. Despite the importance of establishing appropriate sampling protocols, limited research has systematically investigated the minimum number of steps required for reliable measurement of these diverse parameters during outdoor running. This knowledge gap hinders the development of standardized protocols for field-based gait analysis and compromises the quality of research conducted in natural running environments. Therefore, the purpose of this study was to determine the minimum number of steps required to obtain reliable measurements of kinetic, kinematic, gait variability, and $\dot{V}O_2$ parameters during outdoor running with RunScribe™, thereby establishing evidence-based protocols for field-based gait analysis.

Material and methods

Participants

Fourteen runners (2 females/12 males, age 41 ± 16 years, body mass 68 ± 5 kg, stature 1.78 ± 0.06 m, BMI 22 ± 1.7 kg·h^{-2}, training experience 17 ± 8.2 years) voluntarily participated in this study. Inclusion criteria were training volume > 60 km per week, > 3 training sessions per week, wearing the same running gear currently in the last three months, and performing three running race on 10 km in the last six months. A statistical power of 80% ($1-\beta = 0.80$) and $\alpha = 0.05$ were used to determine the necessary sample size of 13 participants. In addition, participants had to be healthy with no neurological or musculoskeletal injuries. This study was approved by the local Ethics Committee institution and conducted in agreement with the local ethical guidelines. After being informed of the procedures, methods, benefits, and possible risks related to the study, each participant reviewed and signed informed consent to participate in accordance with ethical standards.

Experimental design

Testing was carried out in 3 days, interspersed by 5 days in between. On the first day, anthropometric measurements were taken and familiarisation on a track and field with a foot-pods. At the same time, the participant took part in a Critical-Speed test [24]. On the second/third day, as a test–retest, the participant completed 10-minute running trials at their previously assessed individual critical speed, following a 10-minute warm-up at self-selected running speed. All testing sessions were conducted on an official 400-meter outdoor athletics track (0% gradient,

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synthetic surface) under consistent environmental conditions (temperature: 23.4 ± 2.1°C; relative humidity: 15.1 ± 1.8%). Testing occurred between 10:00 and 12:00 to ensure ecological validity while minimizing circadian rhythm effects on performance. Participants wore running clothing and standard training shoes (World Athletics Category A3: traditional training shoes without advanced propulsion technology or energy return systems). For each trial, two sports scientists place side by side each runner with the bikes (Equipped with a GPS Garmin Forerunner Rc730-XT) to monitor the speed as a double speed check.

Data analysis

During each running condition, heart rate (HR) normalized as a percentage of the maximal heart rate (220-age), $\%HR_{max}$) as a proxy for metabolic expenditure was recorded continuously (Cardio Garmin Forerunner Rc730-XT). Participants also reported their rating of perceived exertion (RPE, CR-10 scale) immediately after the end of each run condition.

A foot-pod RunScribe™ system (Scribe Lab. Inc., San Francisco CA, USA) with a sampling rate of 500 Hz (precision of 0.002 s) was also attached to the lace shoe of the right/left leg [25]. The foot pod was calibrated before and re-checked after each trial. For each trial, after the warm-up, a researcher started recording using the RunScribe™ App (Scribe Lab Inc., San Francisco, CA, USA), and each runner was immediately asked to perform a single vertical jump as a start check to provide real-time single-step data on the RunScribe™ App. Results from RunScribe™ were downloaded from their website (https://dashboard.runscribe.com/runs) as.csv files. The following kinematic and kinetic parameters including contact time (CT, ms), flight time (FT, ms), stride length (SL, m), Step Rate (SR, $steps{\cdot}min^{-1}$), leg (k_{leg}) and vertical (k_{vert}) stiffness, Vertical Ground Reaction Force (GRF_v) Rate ($N{\cdot}kg^{-1}{\cdot}s^{-1}$), Horizontal GRF_h Rate ($N{\cdot}kg^{-1}{\cdot}s^{-1}$), and $\dot{V}iO_2$ ($ml{\cdot}kg^{-1}{\cdot}min^{-1}$) were recorded by RunScribe™ App. The phase coordination index (PCI) was calculated from the FT-CT to show the running locomotor variability according to Plotnik et al. [26] and adapted for running [27]. The PCI was determined by adjusting the step time in proportion to the stride time (calculated as the sum of CT (left and right) + FT (left and right) [6]. Here, step time is the duration from the heel strike of one foot to the heel strike of the opposite foot, whereas stride time represents the period between successive heel strikes of the same foot, according to Padulo et al. [6, 10].

For each variable studied, we analyzed data from the initial 600 steps of each 10-minute trial. The 10-minute duration (~1800 total steps at mean cadence) ensured sufficient data beyond the convergence point to verify stability and allowed identification of parameters requiring extended sampling periods up to 900 steps. The convergence method employed in this analysis defines stability as the point where the cumulative running mean remains within ±2% of the final mean value for at least 10 consecutive steps [28–30]. The method ensures that the mean has truly stabilized rather than temporarily passing through the target range. The 2% tolerance threshold represents a practically meaningful difference in biomechanical variables, while the 10-step confirmation window prevents false convergence detection due to random fluctuations [28–30].

Statistical analysis

Results are expressed as mean ± standard deviation between first step till over 600 steps, and were calculated after verifying the assumption of normality of the dependent variables' distributions using the Shapiro–Wilk test. *t*-student test was used to assess differences for $\dot{V}iO_2$, HR, RPE, CT, FT, SR, SL, PCI, k_{leg}, k_{vert}, GRF_v, and GRF_h over the test and retest running conditions. Finally, the reliability of the procedure was evaluated by the intra-class correlation coefficient (ICC) computed with a two-way random model (consistency). Interpretation of ICC value was made according to the scale described by Koo et al. [31] (i.e. poor reliability <0.50; moderate reliability ≥0.50 <0.75; good reliability ≥0.75 < 0.90; and excellent reliability ≥0.90). The significance level was fixed $P \leq .05$ using Statistical Package for Social Science software (version 15.0, IBM SPSS Statistics, Chicago, IL, USA).

Results

All participants concluded all sessions, reaching a critical speed of 14.45 ± 1.39 $km{\cdot}h^{-1}$ (min-max 12.00–16.20 $km{\cdot}h^{-1}$) with an HR of 92.04 ± 5.98 $\%HR_{max}$ and RPE 7.40 ± 0.70 a.u. (wind speeds were 0.2–0-3 $m{\cdot}s^{-1}$ for all running sessions).

For all variables studied, the Shapiro–Wilk test indicated a normal distribution ($P > .05$) for all variables except for GRF_h. Table 1 presents the average/SD with min/max value obtained during the test (steps 1367 ± 74) and re-test (steps 1386 ± 67) related to the first and second 10′ running test. The reliability (Table 1) of the measurements (ICC) was moderate for GRF_h (0.708), and RPE (0.522). A good ICC (Table 1) was found for, SL (0.830), CT (0.878), FT (0.823), PCI (0.873), k_{leg} (0.752), k_{vert} (0.801), GRF_h (0.708), GRF_v (0.872). While an excellent reliability (Table 1) was found on SR (0.928), $\dot{V}iO_2$ (0.942), and HR (0.982).

Table 2 presents the mean ± SD along with the minimum and maximum values for the steps required to

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Table 1. Kinematic/kinetic, PCI, $\dot{V}O_2$, HR, and RPE for both trials.

Variables	Test	Min-Max	Re-Test	Min-Max	ICC	*P*
SR (steps·min^{-1})	176.9 ± 2.7	171.8–180.3	176.7 ± 2.6	172.3–179.8	0.928	.627
SL (m)	2.71 ± 0.15	2.50–2.91	2.70 ± 0.14	2.51–2.90	0.830	.341
CT (ms)	221.3 ± 5.8	213.5–232.4	221.7 ± 5.4	215.1–230.8	0.878	.249
FT (ms)	69.5 ± 3.8	63.7–74.8	69.4 ± 3.6	64.3–74.2	0.823	.148
PCI (%)	2.32 ± 0.46	1.80–3.54	2.57 ± 0.50	1.77–3.71	0.873	.153
k_{leg} (kN·m^{-1})	15.2 ± 0.9	13.9–16.5	15.4 ± 0.9	14.1–16.8	0.752	.326
k_{vert} (kN·m^{-1})	24.5 ± 1.4	22.1–26.3	24.8 ± 1.4	22.4–26.6	0.801	.274
GRF_h (N·kg^{-1}·s^{-1})	8.1 ± 0.4	7.4–8.8	8.2 ± 0.4	7.5–8.9	0.708	.390
GRF_v (N·kg^{-1}·s^{-1})	41.9 ± 1.6	38.9–44.1	42.2 ± 1.6	39.2–44.5	0.872	.196
$\dot{V}O_2$ (ml·kg^{-1}·min^{-1})	47.5 ± 2.3	43.6–50.7	47.8 ± 2.3	43.9–51.0	0.942	.822
HR (%HR_{max})	89.27 ± 7.91	59–95	92.58 ± 4.99	64–97	0.982	.119
RPE (a.u.)	7.90 ± 0.57	7.00–9.00	7.70 ± 0.68	7.00–9.00	0.522	.343

Note: *t*-student (*P* value).

Table 2. Steps for kinematic/ kinetic, PCI, $\dot{V}O_2$ for both trials.

Variables	Test	Min-Max	Re-Test	Min-Max	ICC	*P*
SR (n)	124 ± 36	75–189	119 ± 34	72–178	0.979	.890
SL (n)	152 ± 48	85–245	148 ± 46	82–238	0.994	.895
CT (n)	198 ± 52	125–312	193 ± 50	118–298	0.982	.387
FT (n)	258 ± 68	178–425	278 ± 65	165–412	0.969	.949
PCI (n)	246 ± 12	235–270	247 ± 16	226–272	0.690	.791
k_{leg} (n)	342 ± 78	215–498	326 ± 75	198–485	0.984	.366
k_{vert} (n)	298 ± 71	185–445	291 ± 69	172–428	0.983	.460
GRF_h (n)	168 ± 49	98–258	162 ± 47	92–245	0.989	.233
GRF_v (n)	215 ± 54	142–318	298 ± 52	135–502	0.971	.637
$\dot{V}O_2$ (n)	108 ± 32	65–165	103 ± 30	62–158	0.948	.690

Note: *t*-student (*P* value).

reach the absolute value for kinematic and kinetic variables, PCI, and $\dot{V}O_2$. The reliability of the measurements (ICC), also shown in Table 2, was excellent (>.900) for most of the variables studied, except for PCI, which showed a moderate ICC (0.690). The convergence about the optimal step numbers we find a good agreement within <400 m for $\dot{V}O_2$ (285 ± 84 m) and SR (328 ± 95 m), while in the range 400–600 m fit SL 405 ± 127 m, GRF_h 446 ± 130 m, CT 528 ± 138 m, GRF_v 572 ± 143 m. Conversely, the rest of the variables need more steps as PCI (721 ± 23 m), FT 761 ± 180 m, k_{leg}, 915 ± 207 m, and k_{vert} 796 ± 189 m.

Discussion

The primary objective of this study was to determine the minimum number of steps required to obtain reliable measurements of kinetic, kinematic, gait variability, and $\dot{V}O_2$ parameters during outdoor running with RunScribe™ technology. It is important to clarify that the observed convergence patterns reflect the biological stabilization time required for runners' gait patterns to reach representative steady-state behaviour, rather than technical limitations of the RunScribe™ measurement system. The device's 500 Hz sampling rate and immediate data processing provide more than adequate temporal resolution for instantaneous measurement. The convergence requirements identified represent the inherent variability in human locomotion during outdoor running (Fig. 1). The results successfully address this objective by establishing parameter-specific minimum step requirements that range from 108–342 steps, with important implications for field-based gait analysis protocols.

Parameter-specific convergence patterns

The convergence analysis revealed distinct patterns across different biomechanical domains, directly addressing the study's central research question. Metabolic parameters demonstrated the fastest convergence, with $\dot{V}O_2$ requiring only 108 ± 32 steps (Fig. 1j) equivalent to 285 ± 84 meters to reach reliable measurements at the individual critical speed (14.45 ± 1.39 km·h^{-1}). This finding is particularly significant for practical applications, as metabolic assessment is often a primary outcome in running

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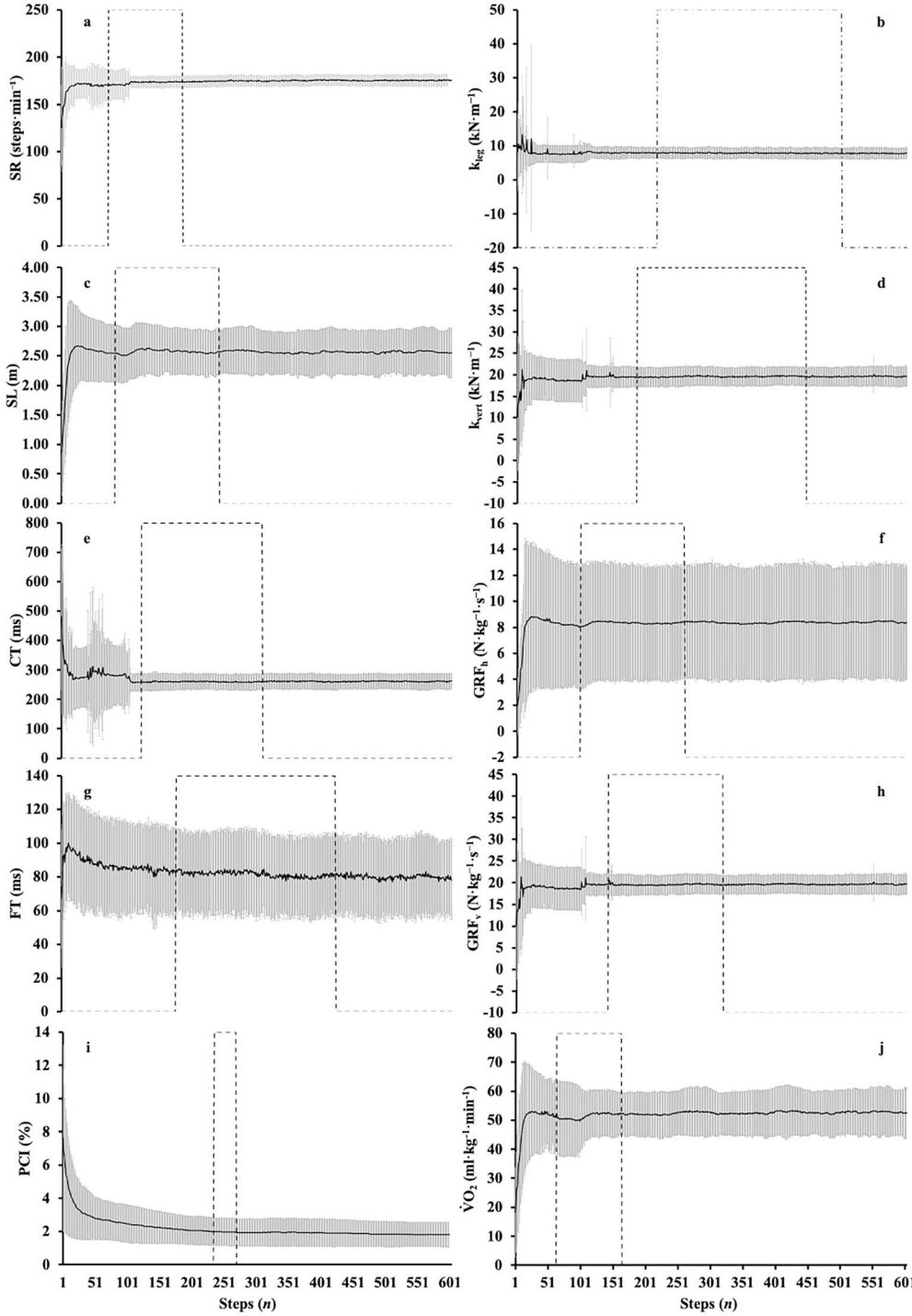

Figure 1. Physiological and biomechanical parameters during prolonged running exercise. Time series data showing (a) step rate, (b) leg stiffness, (c) stride length, (d) vertical stiffness, (e) contact time, (f) horizontal ground reaction force, (g) flight time, (h) vertical ground reaction force, (i) phase coordination index, and (j) oxygen consumption across 600 steps of running. Solid black lines represent mean values, gray shaded areas indicate standard deviation, and dotted lines delineate the range of convergent steps.

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performance research [11, 32]. The rapid stabilization of ViO_2 estimates likely reflects the RunScribe™ algorithm's integration of multiple kinetic variables, which provides a robust composite measure that converges quickly to stable values, consistent with previous observations of composite physiological measures [12]. Step rate (Fig. 1a) showed similarly rapid convergence (124 ± 36 steps, equivalent to 328 ± 95 meters). However, this rapid convergence of step rate contrasts with the slower convergence of contact time (198 ± 52 steps, Fig. 1e), despite their mathematical relationship (step rate = 60/(contact time + flight time in sec)). This apparent discrepancy can be explained by their different computational bases and functional roles: step rate represents the overall frequency of the complete gait cycle, which is tightly regulated as a primary control variable in running. In contrast, contact time represents only the stance phase duration, which shows greater variability as it can be modulated independently through different force production strategies and is more sensitive to terrain micro-variations [15]. This result supports the use of step rate as a reliable parameter for short-duration outdoor assessments and confirms its utility as a primary metric in field-based running analysis, as demonstrated in previous uphill running studies [7, 33].

Kinematic parameters (Fig. 1) displayed moderate convergence requirements, with stride length requiring 152 ± 48 steps (Fig. 1c) 405 ± 127 meters and contact time needing 198 ± 52 steps (Fig. 1e) 528 ± 138 meters, while 258 ± 68 steps for FT (Fig. 1g) equivalent to 761 ± 180 meters. These findings suggest that spatial temporal variables require more extensive sampling than metabolic measures to achieve reliable outdoor measurements [14].

The increased variability in kinematic parameters during outdoor running reflects the influence of subtle surface characteristics (minor texture variations in the synthetic track surface, transitions at lane markings, slight irregularities from track settling and wear, non-uniform force distribution across the surface), variable ambient conditions, and natural stride adjustments [21]. While these variations are minimal compared to natural terrain, they exceed the perfectly uniform conditions of laboratory treadmills [20]. Kinetic parameters showed the most variable convergence patterns. While horizontal ground reaction force (GRF_h) converged relatively quickly at 168 ± 49 steps (Fig. 1f) equivalent to 446 ± 130 meters, vertical ground reaction force (GRF_v) required substantially more steps (215 ± 54 steps, equivalent to 572 ± 143 meters [Fig. 1h]). This discrepancy reflects the different mechanical demands and control strategies associated with propulsive versus impact forces during outdoor running, where surface variations and environmental factors differentially affect these force components [18], consistent with previous observations of force variability in natural environments [30].

Stiffness parameters and gait variability

The study revealed (Fig. 1b and d) that leg stiffness (k_{leg}) required the highest number of steps for convergence (342 ± 78 steps, equivalent to 915 ± 207 meters), followed by vertical (k_{vert}) stiffness (298 ± 71 steps, equivalent to 796 ± 189 meters). This finding is particularly important as it challenges current practices in field-based biomechanical assessment. The extended sampling requirements for stiffness parameters likely reflect their computational complexity, as they are derived from multiple primary variables (ground reaction forces, displacement, contact time) that each carry their own measurement uncertainty [34]. The accumulation of these uncertainties explain why composite variables like stiffness require more extensive data collection for reliable assessment, supporting previous methodological concerns about composite biomechanical measures [28, 29]. The PCI presented unique challenges, requiring 246 ± 12 steps (Fig. 1i) but achieving only moderate reliability (ICC = 0.690). This finding suggests that gait variability measures be inherently more sensitive to environmental perturbations during outdoor running, questioning their utility for short-duration field assessments [26, 27]. The lower reliability of PCI compared to other parameters reflect the increased natural variability in bilateral coordination during outdoor running, where environmental factors continuously challenge the locomotor system [1, 9, 35]. This aligns with previous observations that gait variability measures are particularly sensitive to environmental conditions and require extended sampling periods for reliable assessment [6, 23, 36].

Reliability and practical implications

The excellent reliability achieved for most parameters (ICC > 0.900) validates the convergence methodology and supports the practical application of these minimum step requirements [29, 30]. The finding that different parameter categories require vastly different sampling distances (from ~ 150 m for ViO_2 to ~ 900 m for leg stiffness at typical running speeds) highlights the importance of tailoring measurement protocols to specific biomechanical variables [14, 16]. While the specific step requirements reported here are validated for RunScribe™ technology and cannot be directly extrapolated to other systems without validation, the broader methodological principle—that different biomechanical parameter types require substantially different sampling durations even with the same measurement system—likely applies to other IMU-based

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wearable technologies and field assessment protocols. The hierarchy of convergence rates (metabolic > temporal > kinematic > composite kinetic) reflects fundamental characteristics of locomotor control and variability. Researchers using alternative wearable systems should conduct similar convergence validation analyses to establish system-specific and parameter-specific sampling requirements. These results directly address the fundamental challenge identified in the study objective: establishing evidence-based protocols for field-based gait analysis [17]. The parameter-specific minimum requirements enable researchers to optimize data collection efficiency by tailoring sampling duration to their specific outcome measures of interest, rather than applying arbitrary or laboratory-derived protocols [15, 28].

Environmental and methodological considerations

The outdoor track environment, while providing ecological validity, likely represents a compromise between controlled laboratory conditions and natural running environments [14, 17]. The relatively stable surface and predictable conditions have resulted in faster convergence than would be observed in more variable outdoor settings (trails, roads, varying gradients), as suggested by previous studies examining environmental effects on biomechanical variability [21, 22]. This suggests that the established minimum step requirements represent lower bounds for outdoor running assessment, with more challenging environments potentially requiring extended sampling periods. The use of individual critical speed for standardization represents both a strength and a limitation of the approach. While ensuring comparable physiological stress across participants, the resulting variation in absolute running speeds (12.00–16.20 $km \cdot h^{-1}$) has introduced additional biomechanical variability that influenced convergence patterns, consistent with previous observations of speed-dependent gait characteristics [5, 7].

Conclusion

The study's findings directly support the development of standardized protocols for outdoor gait analysis, addressing a critical gap in sports science methodology [16, 20]. The establishment of parameter-specific minimum step requirements enables more efficient and reliable field-based assessments, potentially expanding the scope and quality of ecological running research [19]. These protocols are particularly valuable for longitudinal studies, training monitoring, and clinical assessments where outdoor measurement validity is paramount [13]. The results suggest that comprehensive biomechanical assessment in outdoor environments requires strategic planning, with researchers needing to balance assessment efficiency against parameter reliability requirements. For studies focusing solely on metabolic or temporal parameters, relatively brief sampling periods (covering ~150–300 m or 1–2 minutes at typical running speeds) may suffice, while comprehensive biomechanical profiling including stiffness measures may require longer data collection (covering ~900 m or 3–5 minutes at typical running speeds of 10–15 $km \cdot h^{-1}$). This aligns with previous recommendations for multi-parameter biomechanical assessment protocols [6, 10, 18].

Author contributions

Johnny Padulo (Validation, Writing—original draft, Writing—review & editing), Raffaele Scurati (Formal Analysis, Methodology, Software), Simone Baldo (Data curation, Investigation, Resources), Fabio Esposito (Visualization), and Leonardo Alexandre Peyré-Tartaruga (Data curation, Formal Analysis, Investigation, Methodology)

Conflict of interest statement

None declared.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Ethics approval

The study received University ethical approval (Ethics Board of the Università degli Studi di Milano), n.79/21. The study was performed in accordance with the ethical standards in the 1964 Declaration of Helsinki and later amendments.

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